

What is statistics?

We may at once admit that any inference from the particular to the general must be attended with some degree of uncertainty, but this is not the same as to admit that such inference cannot be absolutely rigorous, for the nature and degree of the uncertainty may itself be capable of rigorous expression.

— Sir R. A. Fisher

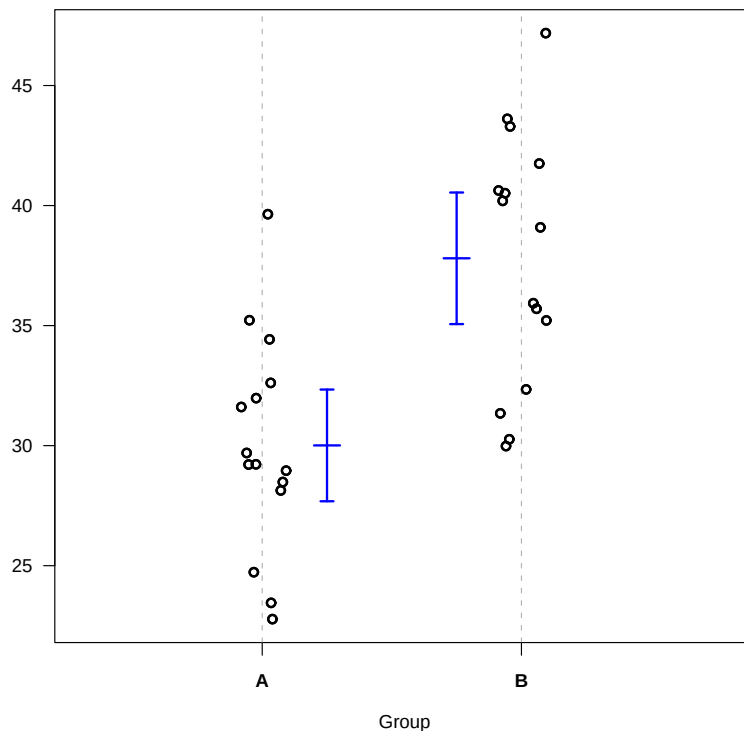
What is statistics?

- Data exploration and analysis
- Inductive inference with probability
- Quantification of uncertainty

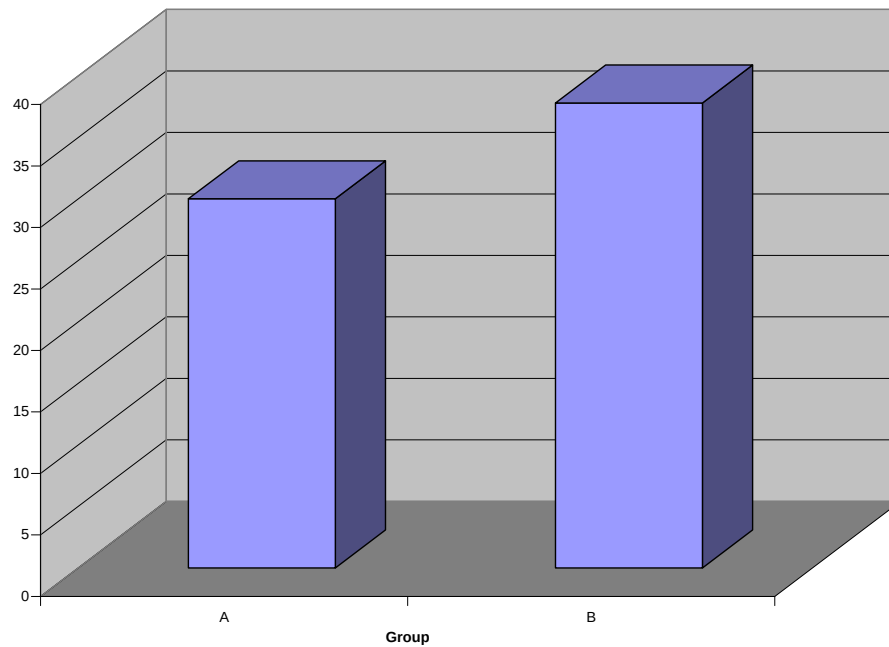
Displaying data

- Show the data
- Avoid chart junk (such as gratuitous 3D)
- Consider taking logs
- In tables, every digit should be meaningful

Good plot



Bad plot



Experimental design

- Randomization
- Replication
- Stratification

Observational studies

Controlled experiment:

The investigator chooses who receives the treatment.

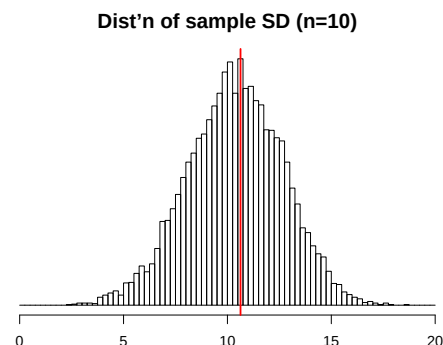
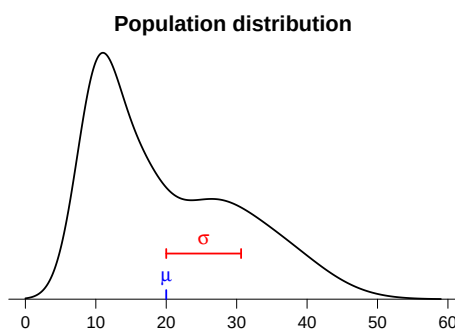
Observational study:

The subjects themselves choose whether they receive the treatment.

Key issues:

- Correlation (association) is not causation.
- Confounding
- Simpson's paradox

Statistical inference



We wish to learn about various features (called **parameters**) of a (possibly hypothetical) population on the basis of sample data.

Estimator:

- A function of the data
- Our best guess of the parameter
- Has a mean, SD (aka SE) and **distribution**

Confidence intervals

A 95% **confidence interval** for a population parameter is an interval indicating the plausible values of the parameter, given the data.

In advance, there is a 95% chance that the confidence interval we obtain will contain the true parameter value.

After the data are obtained, our confidence interval will either contain the true parameter value or it won't—we can no longer talk about chance.

Hypothesis tests

Hypothesis tests (aka tests of significance) are used to answer yes/no questions about population parameters, using data.

- Consider some **null hypothesis**.
- Form a **test statistic** for which large values indicate a departure from the null hypothesis.
- Calculate the **P-value**: the probability of obtaining data as extreme as what was observed, if the null hypothesis were true.
- Interpret the P-value as evidence against the null hypothesis.
- **Don't say**, "The chance that the null hypothesis is true..."

Next term

- Contingency tables
- Analysis of variance (anova)
- More on multiple comparisons
- Two-way anova
- Linear regression
- Multiple regression
- Design of experiments

Example 1

AA	AB	BB
35	43	22

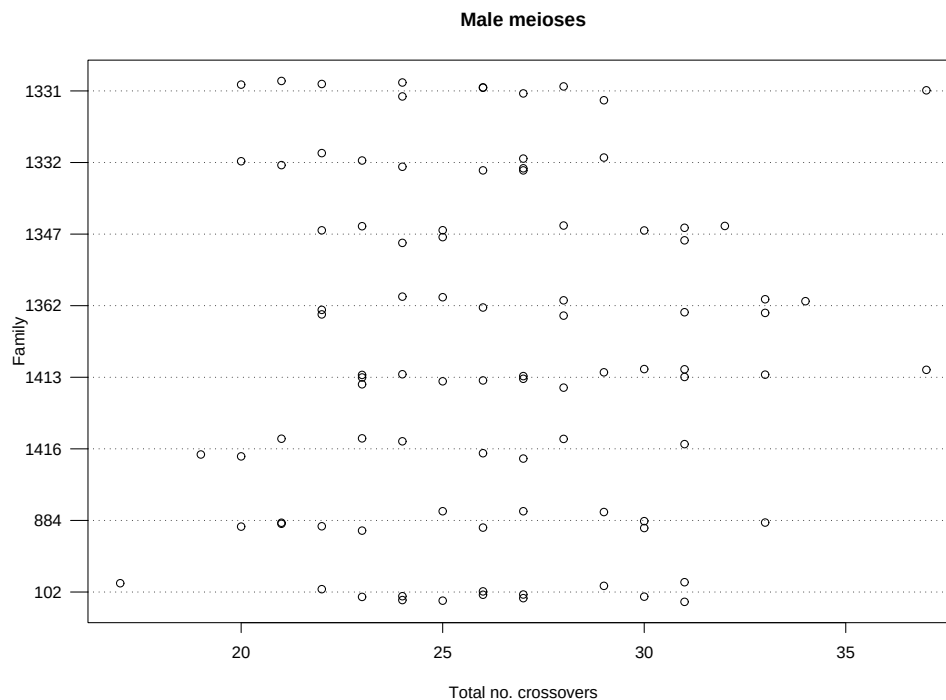
Do these data corresponding reasonably to the proportions 1:2:1?

Example 2

Strain	No. (%) dead	No. mice
A	14 (47)	30
B	9 (90)	10
C	9 (60)	15
D	11 (61)	18
E	16 (80)	20

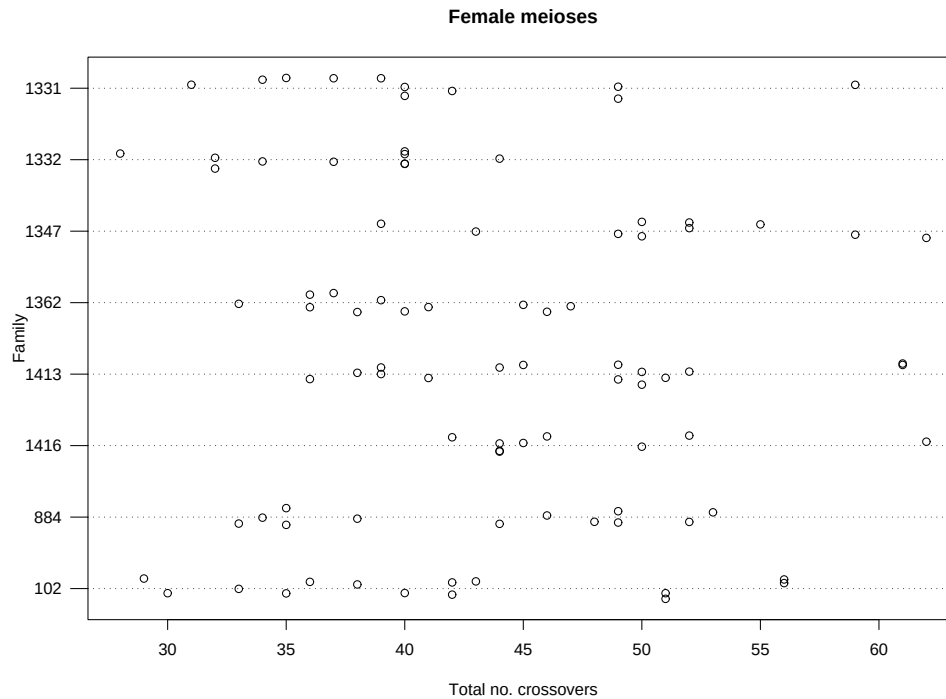
Do the strains differ in their survival rates?

Example 3a



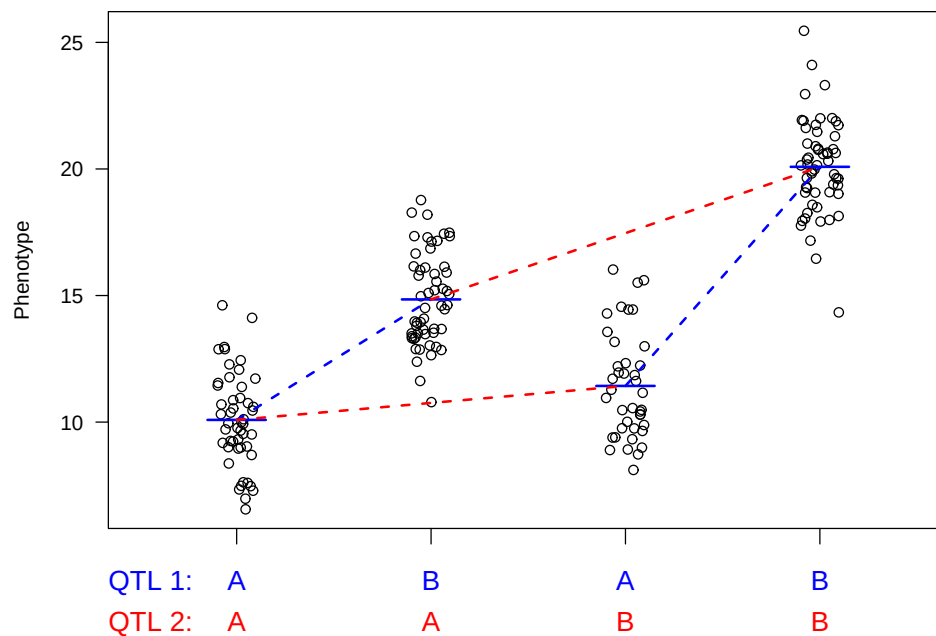
Do the fathers differ?

Example 3b



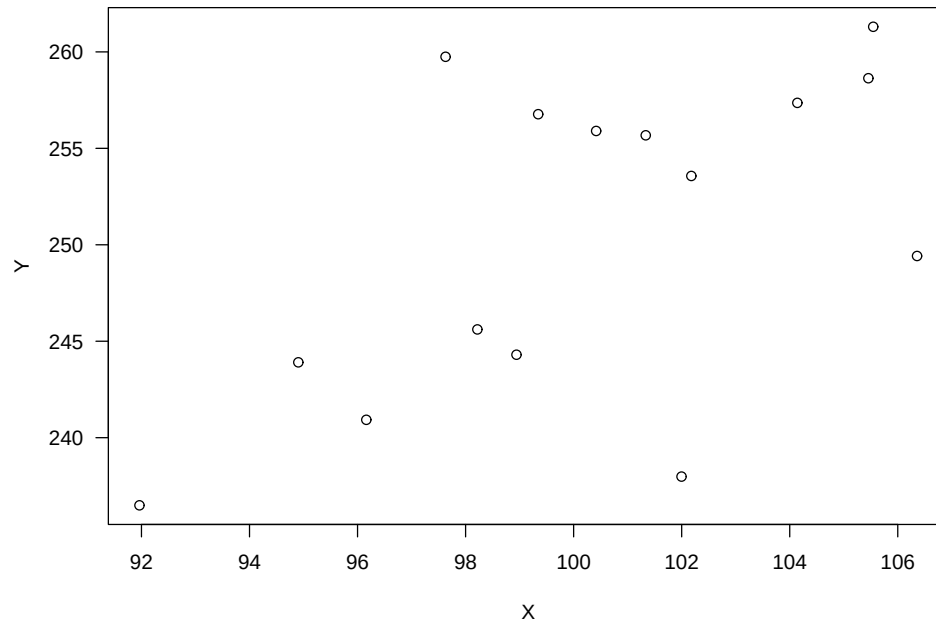
Do the mothers differ?

Example 4



Do these QTLs interact or do they act additively?

Example 5



Are these variables associated?